

# MSME IDEA HACKATHON 6.0

Reference No. :- **26INC06WB070151**

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| <b>1. Business Incubator Name</b>                               | Dr. Subhabrata Banerjee                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>2. Business Incubator Email ID</b>                           | subhabrata.banerjee@iem.edu.in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>3. Business Incubator Mobile Number</b>                      | 9433569949                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>4. HI/BI State</b>                                           | WEST BENGAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>5. HI/BI Name</b>                                            | Institute of Engineering & Management, Kolkata , under University of Engineering and Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>6. Theme</b>                                                 | Other Frontier Technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>7. Idea Sector</b>                                           | Miscellaneous Sector (Environment, Forests, Water & Sanitation; Foods, Beverages, FMCG, Consumer Goods; Infrastructure, Construction, Housing; IT, ITES, Electronics, White Goods, Telecommunication; Metals, Engineering, Machinery, Automation and Transportation, Automotive, E Vehicles, Railways, Aviation, UAV and any other sub-sector)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>8. Title of proposed idea/ innovation</b>                    | ReForge AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>9. Briefly explain newness/ uniqueness of the innovation</b> | Unlike conventional reverse engineering tools that only reconstruct a CAD model from a scanned part, ReForge AI provides an end-to-end AI-powered import substitution platform. It not only recreates the component but also recommends suitable materials, identifies the optimal manufacturing process (CNC, casting, or 3D printing), estimates production cost and lead time, and suggests capable local MSMEs for manufacturing. This transforms reverse engineering into an intelligent manufacturing decision-support system, enabling affordable indigenous production of imported spare parts and reducing dependence on foreign suppliers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>10. Concept &amp; Objective</b>                              | ReForge AI is an AI-powered Industrial Reverse Engineering and Manufacturing Intelligence Platform designed to help MSMEs locally reproduce imported, obsolete, or unavailable industrial spare parts. Users upload photographs or a 3D scan of a damaged component along with basic dimensions, and the platform uses computer vision and generative AI to reconstruct an accurate CAD model. It then recommends the most suitable material, manufacturing process (CNC machining, casting, forging, or 3D printing), estimates production cost and lead time, and identifies capable local MSMEs for manufacturing. The objective is to reduce India's dependence on imported industrial components by enabling fast, cost-effective, and indigenous manufacturing. Beyond reverse engineering, ReForge AI acts as a manufacturing decision-support system by optimizing production methods, suggesting design modifications for manufacturability and cost reduction, and helping industries quickly replace obsolete or discontinued parts. The platform aims to strengthen the MSME ecosystem, promote import substitution, reduce machine downtime, preserve manufacturing continuity, and support the vision of Atmanirbhar Bharat through AI-driven innovation in precision manufacturing. |

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| <b>11. Specify the potential areas of application in industry/ market in brief</b> | <p>ReForge AI can be deployed across a wide range of manufacturing sectors that rely on imported, obsolete, or custom-engineered components. Its primary applications include automotive, aerospace, railways, defence, heavy engineering, machine tools, oil gas, power generation, mining, pharmaceuticals, food processing, textiles, and industrial automation. The platform is particularly valuable for MSMEs that face production delays due to unavailable spare parts, discontinued components, or long import lead times. Maintenance, Repair, and Overhaul (MRO) companies can use ReForge AI to rapidly recreate damaged components, while OEMs can accelerate product development and localize their supply chains. Tool rooms, precision machining workshops, foundries, and 3D printing service providers can leverage AI-generated CAD models and manufacturing recommendations to reduce design time and improve production efficiency. The platform also supports government initiatives such as Atmanirbhar Bharat and Make in India by enabling import substitution, strengthening indigenous manufacturing capabilities, and creating opportunities for local MSMEs to participate in high-value manufacturing. With applications in reverse engineering, spare-part localization, rapid prototyping, manufacturing process planning, and supply-chain resilience, ReForge AI has strong commercialization potential across India's industrial ecosystem.</p> |
| <b>12. Briefly provide the market potential of idea/ innovation</b>                | <p>India imports a significant number of industrial spare parts, precision components, and machine assemblies, resulting in high procurement costs, long lead times, and production downtime for MSMEs. ReForge AI addresses this gap by enabling affordable, AI-assisted reverse engineering and indigenous manufacturing. The primary market includes over 63 million MSMEs in India, along with OEMs, MRO service providers, machine tool manufacturers, foundries, precision engineering firms, and 3D printing service bureaus. The platform can be offered as a Software-as-a-Service (SaaS) solution with subscription plans, pay-per-project reverse engineering, and enterprise licensing. Additional revenue opportunities include AI-powered manufacturing consulting, CAD generation services, and partnerships with local manufacturing networks. As industries increasingly adopt digital manufacturing and pursue supply-chain localization under initiatives like Atmanirbhar Bharat and Make in India, ReForge AI has strong commercial potential. The platform can also be expanded to international markets where manufacturers seek cost-effective reverse engineering and localized production solutions, making it a scalable and globally relevant innovation.</p>                                                                                                                                                                                          |
| <b>13. Uploaded Block diagram/ flow chart/ Circuit Diagram/ Pictures</b>           | <p>View/Download</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |